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FLOOD INUNDATION RISK MODELING

Technical Report to the Director of Disaster Planning

Summary: Calgary model AUC 0.912 correctly identifies 70% of flood zones. Deployed to Minneapolis: 6.2% of the city (1,146 cells) flagged as flood-prone, concentrated along the Mississippi River gorge, Minnehaha Creek, and the Chain of Lakes corridor.

1. Introduction

This report develops and validates a binomial logistic regression model of flood inundation risk using the Calgary, Alberta 2013 flood as the training event. In June 2013, sustained precipitation caused the Bow and Elbow rivers to crest at record levels, inundating thousands of hectares of urban and peri-urban land, displacing over 100,000 residents, and producing an estimated $6 billion CAD in damages, the costliest natural disaster in Canadian history.

The modeling approach works as follows. The Calgary study area is divided into a regular 60 m fishnet grid. Each cell is assigned a binary outcome, flooded or not flooded, derived from the satellite inundation raster, and a set of six engineered features drawn from terrain analysis, remote sensing, and hydrographic data. A logistic regression model is then fit to 70% of cells and evaluated on the held out 30%. Because flood events are rare relative to the total urban land area (prevalence approximately 5.1%), the decision threshold is set to 0.10 rather than the conventional 0.50, prioritizing sensitivity over specificity in keeping with the asymmetric costs of missing actual flood zones in an emergency planning context.

The model performs well on the Calgary test set, achieving an AUC of 0.912 and a sensitivity of 70% at the 0.10 threshold. These results support deployment to Minneapolis, Minnesota, a comparable riverine city at the confluence of the Mississippi and Minnesota rivers, where the Calgary-trained model generates a spatially coherent flood risk surface despite having no Minneapolis-specific outcome data.

2. Features

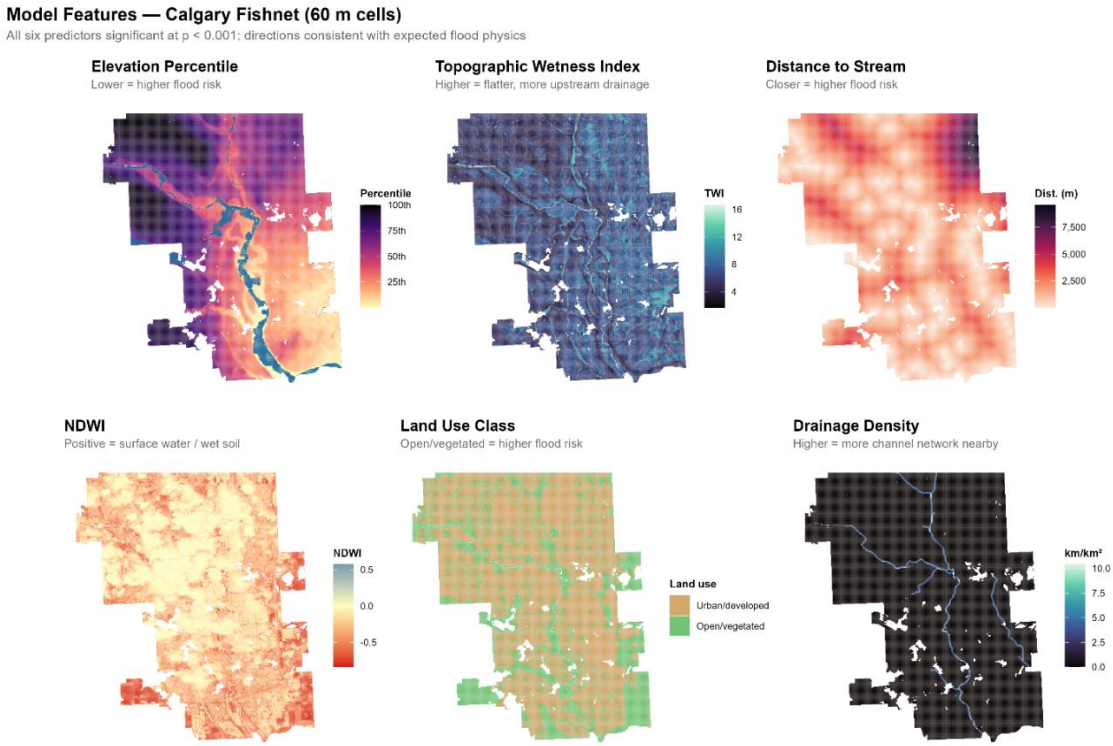
Seven candidate features were engineered from freely available spatial data. Three were dropped during model development: slope (redundant once TWI is included), distance to lakes (non-significant in the presence of drainage density), and Curve Number (spatial confound: high-CN urban cells sit on elevated ground away from river valleys). The six retained features are described in Table 1.

Table 1. Features included in modeling with source, expected direction, and inclusion rationale

Feature	Source	Expected Direction	Rationale
Elevation percentile	Calgary DEM (ADF); SRTM for Minneapolis	Negative	Low-lying cells concentrate flood water; expressed as city-wide percentile rank for cross-city transferability.
TWI (watershed feature)	Computed from filled DEM via Whitebox	Positive	Topographic Wetness Index = $ln\left(\frac{flow_{acc}}{tan(slope)}\right)$ High TWI = flat cell with large upstream drainage area, the floodplain signature.
Distance to stream	Calgary Open Data hydrology CSV; Minneapolis Water shapefile	Negative	Cells closer to water courses are more likely to be inundated by overbank flow. Units: meters.
NDWI	Sentinel-2 Bands 3 and 8 (Green, NIR)	Positive	Normalized Difference Water Index. Cells with pre-event surface moisture are more flood prone.
Land use class	Derived from NDWI: open/vegetated (3) vs. urban.	Positive	Open and vegetated riverside land is more flood-prone than elevated urban fabric. Replaced Curve Number after CN showed a spatial confound.
Drainage density	Computed: km stream channel per km² of cell area	Positive	Higher channelization density increases runoff concentration; subsumed flow accumulation in multivariate tests.

Elevation percentile was derived from the Calgary DEM, reprojected to UTM Zone 11N, and aggregated to the 60 m fishnet via zonal mean. Raw elevation was converted to a within-city percentile rank (0–100), which allows the feature to transfer to Minneapolis without recalibration regardless of absolute altitude differences between the two cities. **Topographic Wetness Index (TWI)** was computed from a depression-filled DEM using D8 flow accumulation and local slope: TWI = ln(flow accumulation

/ tan(slope)). It serves as the watershed feature, capturing how much water a cell tends to accumulate relative to how quickly it drains. High values mark flat, well-connected valley floors, the floodplain signature. **Distance to the nearest stream** measures Euclidean distance in meters from each fishnet cell centroid to the nearest water course, using Calgary's 415 open-data hydrology polygons. It directly captures proximity to the primary flood source. **Normalized Difference Water Index (NDWI)** was computed from Sentinel-2 Green and NIR bands as (Green − NIR) / (Green + NIR), then aggregated to the fishnet via zonal mean. It captures pre-event surface moisture conditions, with positive values indicating surface water or wet soil at the time of the satellite overpass. **Land use class** collapses Calgary's land use districts into a binary variable: urban/developed versus open/vegetated. If any urban polygon overlapped a fishnet cell, that cell was assigned urban, a conservative rule reflecting that even small impervious patches dominate runoff. **Drainage density** measures kilometers of stream channel per km² within a 660 m neighborhood around each cell, derived entirely from the flow accumulation raster. It captures local channelization intensity and transfers to Minneapolis using only the DEM, requiring no additional hydrographic data.



3. Model – Logistic Regression & Predictions

A binomial logistic regression was fit to predict the probability of flood inundation for each 60 m fishnet cell. Logistic regression was chosen for its interpretability: each coefficient directly expresses the change in log-odds of inundation associated with a one-unit increase in the predictor, and the model's parametric form allows it to be applied to Minneapolis without retraining.

The Calgary fishnet was partitioned into a 70% training set (137,789 cells; 6,958 flooded) and a 30% test set (59,052 cells; 3,059 flooded) using stratified random sampling on the flood outcome, preserving the approximately 5% flood prevalence in both partitions. The model was fit exclusively on the training set and all performance metrics reported below are computed on the held-out test set, ensuring unbiased estimates of out-of-sample accuracy. Because flood cells represent only about 5% of the study area, the decision threshold was set to 0.10 rather than the conventional 0.50. At 5% prevalence, a classifier that predicts "not flooded" for every cell achieves 95% overall accuracy while identifying zero actual flood zones. The lower threshold shifts the sensitivity-specificity tradeoff toward capturing a

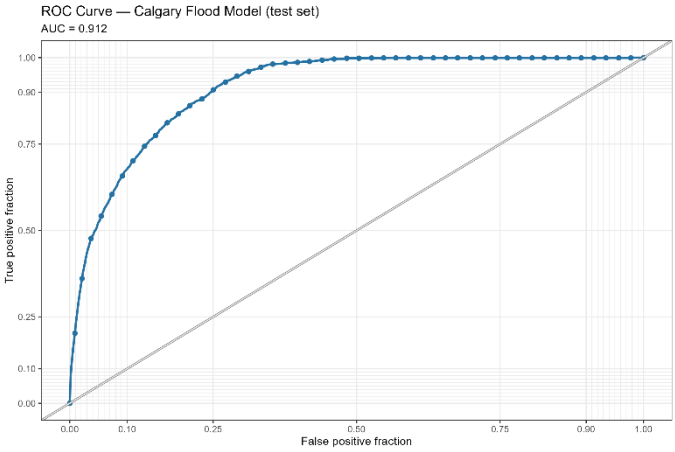
higher share of true flood cells at the cost of accepting more false positives, which is appropriate when missing a flood zone carries substantially higher planning costs than a false alarm.

Table 2 presents the model coefficients. All six predictors are significant at $p < 0.001$ and behave in physically interpretable directions. Elevation and distance to stream are negative, meaning flood probability decreases as cells sit higher or farther from channels. TWI, NDWI, open/vegetated land use, and drainage density are all positive. NDWI carries the largest odds ratio at 7.92, reflecting that cells already displaying surface moisture at the time of the Sentinel-2 overpass are overwhelmingly likely to fall within the flood boundary.

Table 2. Logistic regression model, Calgary flood inundation.

Term	Log-Odds	Std. Error	Odds Ratio
(Intercept)	-0.4181	0.0661	0.6583
Elevation percentile	-0.0448	0.0008	0.9562
TWI	0.0527	0.0067	1.0541
Distance to stream	-0.0013	0.0000	0.9987
NDWI	2.0699	0.0734	7.9240
Land use class (open/veg.)	1.0121	0.0308	2.7513
Drainage density	0.3951	0.0109	1.4846

The ROC curve above (Fig. 1) plots the tradeoff between true positive rate and false positive rate across all possible thresholds. A curve bowing toward the top-left corner indicates the model discriminates well between flooded and non-flooded cells across the full probability range, well above the diagonal that would represent a random classifier. The model achieves an AUC of 0.912, meaning it correctly ranks a randomly drawn flooded cell above a randomly drawn non-flooded cell 91.2% of the time. The McFadden pseudo-R^2 of 0.347 indicates good overall model fit for a binary outcome.



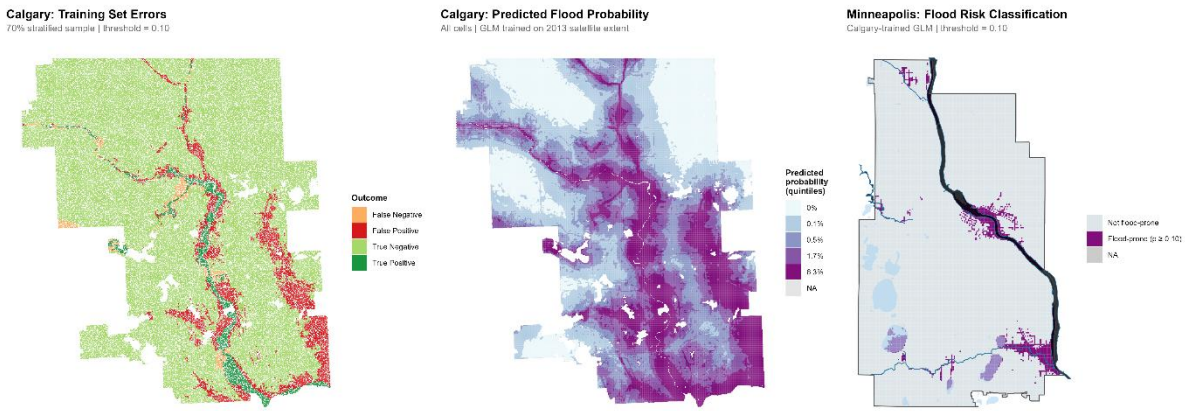
At the 0.10 threshold, the model achieves a sensitivity of 69.9% and a specificity of 89.1% on the test set. Table 3 describes the four classification outcomes in plain terms, with counts from the test set.

Table 3. Confusion matrix outcomes by count and interpretation

Outcome	Predicted	Observed	Count	Interpretation
True Positive	Flooded	Flooded	2,137	Correctly flagged as flood prone. Priority target for infrastructure protection.
True Negative	Not flooded	Not flooded	49,909	Correctly cleared. No mitigation resources are directed here.
False Positive	Flooded	Not flooded	6,084	Over-predicted as flood prone. May trigger unneeded mitigation investment.
False Negative	Not flooded	Flooded	922	A missed flood zone. The most dangerous error type for public safety.

The three maps below show the spatial distribution of these outcomes. The training set confusion map reveals that true positives trace the Bow and Elbow river corridors tightly, while false negatives are scattered rather than clustered, suggesting no systematic spatial gap in the feature set. False positives form broader bands adjacent to the channels, a direct consequence of the low threshold. The full

Calgary prediction map shows flood probability declining sharply with distance from the channel network, with the highest-risk quintile concentrated in the valley floors. The Minneapolis map shows the Calgary-trained model flagging 6.2% of the city as flood-prone, with predicted risk concentrated along the Mississippi River gorge on the western boundary and near the Minnesota River confluence to the south, consistent with known flood hazard zones in the city.



4. Conclusion:

The logistic regression model trained on the Calgary 2013 flood event performs well on held-out data. An AUC of 0.912 indicates strong discriminative ability, and a sensitivity of 69.9% at the 0.10 threshold means the model correctly identifies approximately 7 in 10 truly flooded cells. All six predictors behave in physically sensible directions and the spatial error structure shows no systematic clustering of false negatives, suggesting no critical missing feature. The false positive rate of 11% is an acceptable tradeoff in a planning context where missing a flood zone carries substantially higher costs than a false alarm. Outputs should be treated as a screening tool for prioritizing further investigation rather than a definitive hazard boundary.

Deployment to Minneapolis produces spatially coherent results, with predicted risk concentrating along the Mississippi River gorge and the Minnesota River confluence, both well-documented flood hazard zones. Without observed inundation data for Minneapolis, the predictions cannot be validated directly, and this caveat should accompany any operational use of the outputs. More broadly, this approach has genuine utility for preliminary flood hazard screening in data-sparse environments: it requires only freely available inputs, trains quickly, and produces interpretable coefficients that planners can communicate to non-technical audiences. Its principal limitation is that it learns the signature of one event under one set of hydroclimatic conditions and would perform less reliably for floods driven by stormwater overflow or groundwater emergence rather than riverine inundation. It is recommended that we improve the model, by training it on more flooding events or failing to acquire sufficiently parallel data, we could oversample the flooded areas or prune the dataset that has no flood overrepresentation.

5. Sources:

Calgary DEM, Flood Indunation, and study area boundaries were provided by the course. Calgary Hydrology was sourced from Calgary Open Data. Minneapolis DEM is from USGS's Earth Explorer. Whereas City Boundary and Hydrology were from Minneapolis Open Data. Soil Groups were extracted from EarthDATA for Calgary and Minneapolis. Finally, Copernicus Sentinel-2 imagery were downloaded for their B03 and B08 raw captures using Copernicus dataspace.